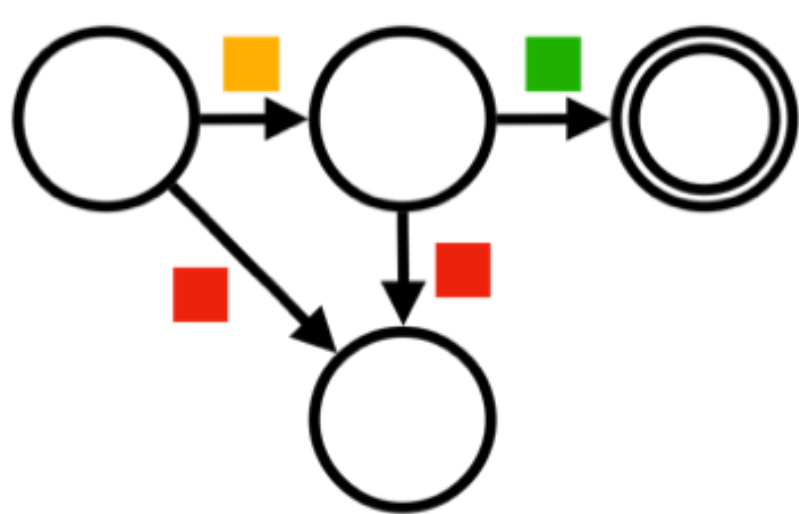


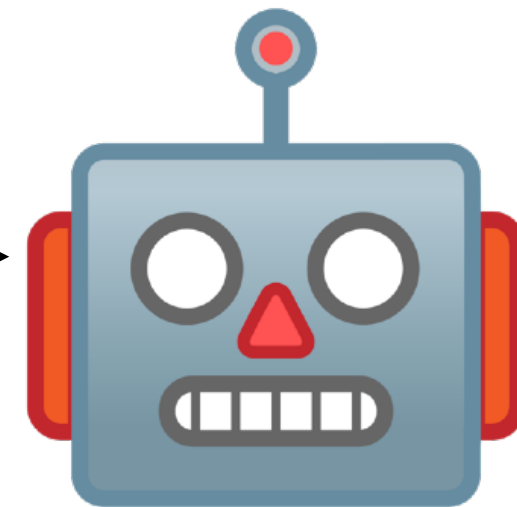
- **Challenge:** How to generate automaton-instruction pairs?

1. Which modality is better for learning-based control?
 2. What happens when these modalities are combined?
- **Challenge:** How to generate automaton-instruction pairs?

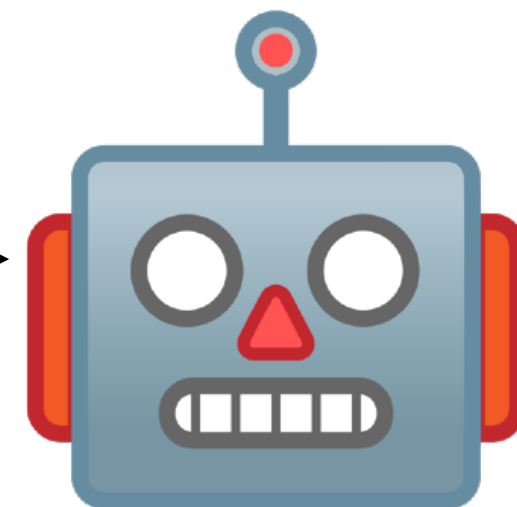
Recap: Our goal is to compare the two modalities.



(Automata)



Pick up the laundry
and fold it. Avoid
breaking anything!



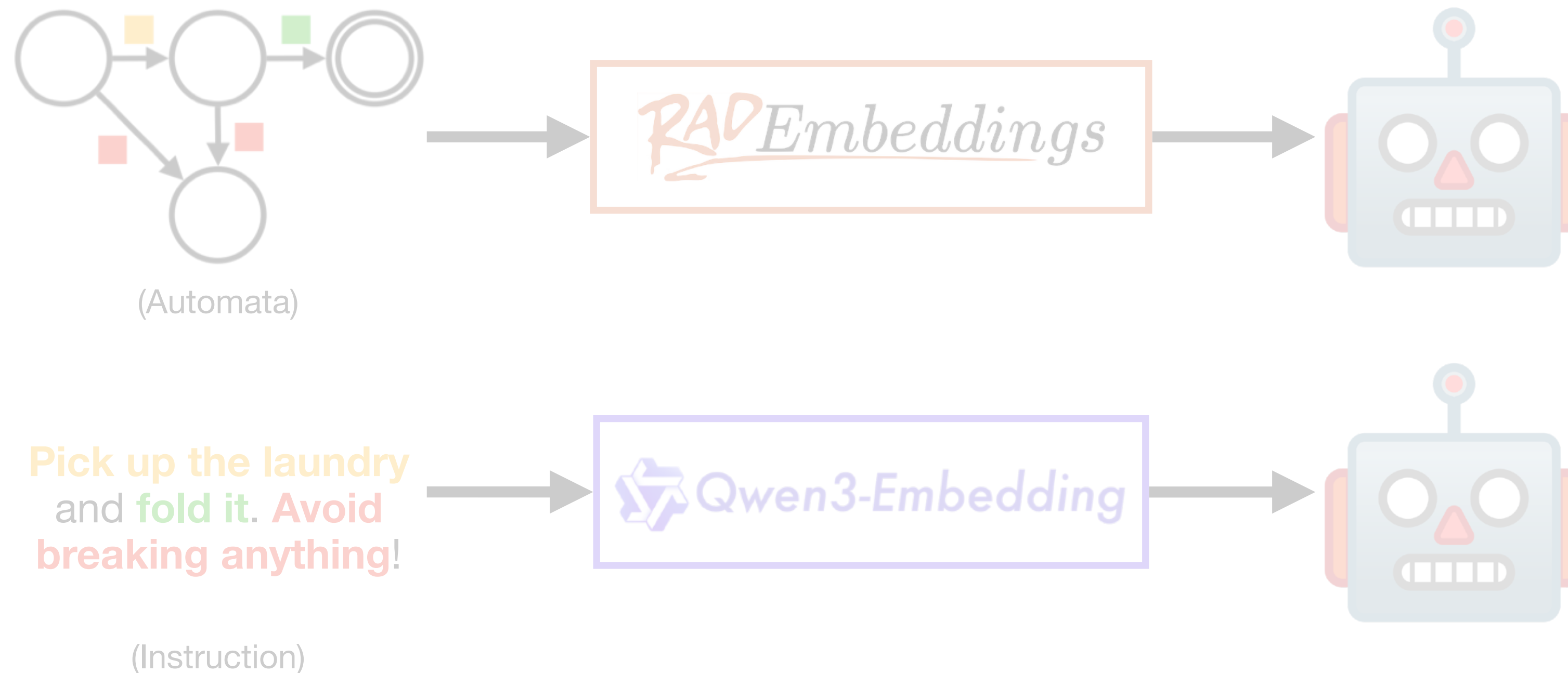
(Instruction)

Yalcinkaya, Beyazit, et al. "Compositional automata embeddings for goal-conditioned reinforcement learning." Neurips 2024.

Zhang, Yanzhao, et al. "Qwen3 embedding: Advancing text embedding and reranking through foundation models." ICMML 2026.



Recap: Our goal is to compare these two modalities.



1. Which modality is better for learning-based control?
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Outline

- ~~Introduction~~
- ~~Preliminaries~~
- **Synthesizing Language Instructions from Automata**
- Learning Policies with Automata and Instruction Embeddings
- Empirical Study
- Discussion